

UCI201: INTERNET AND WORLD WIDE WEB

INTRODUCTION

In 1990 and 1991, Tim Berners-Lee created the World Wide Web at the European Laboratory for Particle Physics (CERN) in Geneva, Switzerland

The original purpose of the World Wide Web (WWW) was to provide easy access to cross-referenced documents that existed on the CERN computer network

The Web technology that we now enjoy is one of humankind's most empowering developments. This technology empowers both the message sender and the message recipient—in other words, the publisher of content and the reader of that content

Many people now turn to the Web to find information in their daily lives, rather than using the telephone or other traditional means.

The Web allows information to be disseminated with speed, accuracy and detail.

Web addresses are now included in most businesses' radio, television and print ads, offering customers a more personalized and specific method of information access to assist in learning and decision making.

For long-time Web users, the transformation of Web design is clear. Early sites consisted of

lengthy, plain-textual content. This trend evolved into developed design practices that rival

other ad media such as television and print.

To be competitive, people in technical positions (and those who support them) must be familiar with Web design concepts. Everyone in the modern office environment will contribute to the development of the information infrastructure.

Tools and technologies abound to make Web page design easier. Designers must choose the

tools that will give them the competitive edge for their organizations.

HISTORY OF THE INTERNET

Mid 1960: Papers on "Packet Switching" emerge.

End 1969: ARPA (Advanced Research Projects Agency) sponsors the development of a packet-switching network, called the ARPANET.

1974: The TCP/IP protocols and model are being proposed by Cerf/Kahn.

1980: IPv4 is introduced

1983: ARPANET adopts TCP/IP. At this time, the ARPANET has 200 routers.

1984: NSF (National Science Foundation) funds a TCP/IP based backbone network. This

backbone grows into the NSFNET, which becomes the successor of the ARPANET.

1995: NSF stops funding of NSFNET. The Internet is completely commercial.

The Internet originated in the late 1960s when the United States Defense Department developed ARPAnet (Advanced Research Projects Agency network), an experimental network of computers designed to withstand partial outages such as a bomb attack. The

agency sought to create a web of computers that could continue to communicate with e

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other, even if some of the computers were disabled. In the mid-1980s, when desktop computer workstations became increasingly popular, organizations wanted to connect their local area networks (LANs) to ARPAnet. If computers could link together and share resources, everyone would benefit.

January 1, 1983 is considered the official birthday of the Internet. Prior to this, the various

computer networks did not have a standard way to communicate with each other. A new communications protocol was established called Transfer Control Protocol/Internetwork Protocol (TCP/IP). This allowed different kinds of computers on different networks to "talk" to each other. ARPANET and the Defense Data Network officially changed to the TCP/IP standard on January 1, 1983, hence the birth of the Internet. All networks could now be connected by a universal language.

To help speed the connections, the National Science Foundation (NSF) established five super computing centers in 1986, creating the NSFnet backbone. In 1987, the NSF signed a

cooperative agreement to manage the NSFnet backbone with Merit Network, Inc., and by 1990, ARPAnet had been phased out. NSFnet continued to grow, and more and more countries around the world connected to this Internet backbone.

1991 was a big year for the Internet: The National Research and Education Network (NREN) was founded and the World Wide Web was released. The Internet is still

dominated by scientists and other academics, but begins to attract public interest. With t

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release of the Mosaic Web browser in 1993 and Netscape in 1994, interest in and use of the

World Wide Web exploded. More and more communities become wired, enabling direct connections to the Internet. In 1995, the U.S. federal government relinquished its management role in the Internet and NSFnet reverted back to being a research network. Interconnected network providers are strong enough now to support U.S. backbone traffic

on the Internet. However, the presidential administration encourages continued development of the U.S. backbone of the Internet, also known as the National Information Infrastructure (NII) and, most commonly, as the "Information Superhighway".

USES OF THE INTERNET

1. Communication

Communication is the most important gift that the internet has given to the common man.

Email, social networking sites are some of the prime examples of it. This is one such gift

of the internet which is cherished by everyone and has made our life easier to much extent.

2. Research

In order to do research you need to go through hundreds of books as well as the references

and that was one of the most difficult jobs to do earlier. You just have to search for the concerned topic and you will get hundreds of references that may be beneficial for your research. And since internet is here to make your research public, you can then benefit a large amount of people from the research work that you have done. Research is one such

thing which has got lots of benefit from this evolution of internet. Research process has now got wings and has gained the most due to the internet

3. Education

There are a number of books, reference books, online help centres, expert's views and other study oriented material on the internet that can make the learning process very easier as well as a fun learning experience. There are various number of tutorials available

over the internet using which you can learn so many thing very easily. There can't be any excellent use of the internet other than education as it is the key to achieve everything in life.

4. Financial Management

The internet service provides us with the great facility which can make your money management well and is called as online banking. Through internet money can be transferred from one place to another easily within few minute. This facility is not just a boon for commercial purposes but is also an amazing thing for all the people since money

is concerned with everyone and so this is used worldwide for the transfer of money.

5. Real Time Updates

This regards to the news and other happenings that may be on-going in different parts of the world. They include updates on business, sports, finance, politics and entertainment.

6. Leisure

This includes watching videos to listening songs, watching movies, playing games, chatting with the loved ones etc.

7. Job Search and Recruitment

Job search is one such thing which required updates from every corner as well the patience from the person searching for it. Using internet, this has become an easier task. Job search has been placed at the number eighth place. There are endless amount of websites on the internet that provide with the news about the vacancy in various companies. You just need to get yourself registered at these website and they will do the rest. They will not only send you the email mentioning job vacancy according to your choice but will also help you to select the best out of them. Job search is easier now, all thanks to internet else it was a nightmare earlier

9. Blogging

The internet has enabled a large number of people not only write blogs but can also publicize their work so that their work reaches to most of the people.

10. Shopping

There is huge number of options available for a common people to buy or to sell any particular item using the internet. Using internet now it is possible to buy products from a

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over the world. Some e-commerce site available include amazon.com, olx.com

2. CONNECTING TO THE INTERNET

Understanding Terminology

- Always on: a feature of Internet connection in which the connection is ready to use immediately as long as the computer has power.
- Analog signals: continuously changing natural signal like human voice. The same technology used to record sounds onto an audiotape.
- Bandwidth: the amount of data that can be transmitted across a network or cable; usually measured in bits per second (bps) for Internet access. For example, a modem may connect at 28.8 Kbps, which means it can nominally send and receive 28,800 bits of information per second.
- Broadband: a high-speed Internet access that offers an always-on connection, which is called in contrast to a dial-up connection using analog modem.
- Digital signals: the signals for the performance of computer by turning on and off a series of electronic switches represented by the numerical digits of 0 (the code for

off) and 1(the code for on). The combinations of these digital codes represent computer text, commands and graphics.

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- Internet service provider (ISP): a company that provides access to the Internet. For a monthly fee, the service provider gives a software package, username, password and access phone number. In addition to serving individuals, this also serves companies, providing a direct connection from the company's networks to

the Internet.

PREREQUISITES

There are five requirements for connecting to the Internet.

1. Service provider

You will normally have to subscribe to a company that will provide you with access to the Internet through their servers.

2. Software

Your computer must have the correct type of software to let it transmit and receive data across the Internet. You will need (at least) a browser to view web pages and e-mail software to send and receive e-mails. There are two main types of this software that we commonly use.

- A browser.

This is a program which lets you view web pages. Most browsers also have file transfer capabilities, as well as letting you use chat services. Examples of browsers include Internet Explorer, Netscape Navigator and Safari.

- An e-mail client

This is a program which sends and receives e-mail, and gives users access to newsgroups. E-mail software also allows the user to organize messages into folders, store commonly used e-mail addresses, send and receive attachments and send e-mails to groups. Examples of e-mail clients include Eudora, Mail and Outlook Express. Most ISPs also allow users to view e-mail directly using a browser (e.g.

hotmail).

- Mobile Apps

3. Hardware

The specification of the computer must be sufficient to run the software – in particular, i

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will need a fast enough processor and enough RAM. Typical minimum hardware requirements are as follows:

	Apple Mac	PC
Processor	266 MHz	233 MHz
Memory	64 Mb RAM	64 Mb RAM
Hard disk space	72 Mb	52 Mb
Operating system	MAC OS 10 +	Windows 98 +

4. Connection

You need to have a telecommunications connection – either a phone line, a cable or satellite connection or a WAP phone. Businesses may also have high speed leased lines to allow multiple users fast access to Internet services.

Type of connection	Speed of connection
Dial-up	56 Kbps
ADSL broadband	Up to 1 Mbps
Cable broadband	Up to 3 Mbps
Satellite broadband	Varies

5. Modem

You need a suitable device to connect your computer to the telephone line, or cable or satellite. Normally, this will be some type of modem.

Types of Connections Available Today

A. Cabled Connections

i) Dial-up Internet Access

This is an internet access service that enables one to access the internet using the normal telephone line and a modem. This service is very useful to people who travel a lot and want a service that is easily available, cost effective so as to access the internet to transmit light data. Once the Consumer is using the service, the telephone line is not available to receive or to make calls.

Typical Dial-up connection speeds are 56 Kbps.

A. Affordable

D. Slow

ii) DSL – Digital Subscriber Line

Frequently referred to as an always on connection. It uses existing 2-wire copper telephone line connected to the premise so service is delivered simultaneously with wired telephone service – it will not tie up your phone line as an analog dial-up connection does.

The two main categories of DSL for home subscribers are called ADSL and SDSL.

1. ADSL - Asymmetric Digital Subscriber Line

ADSL is the most commonly deployed types of DSL. Short for asymmetric digital subscriber

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line ADSL supports data rates of from 1.5 to 9 Mbps when receiving data (known as the downstream rate) and from 16 to 640 Kbps when sending data (known as the upstream rate). ADSL requires a special ADSL modem.

Connection speeds – Sending data 16 - 640 Kbps

Connection speeds – Receiving data 1.5 – 9 Mbps

2. ADSL+2 ?- ADSL ?Extension

An extension to ADSL broadband technology that provides subscribers with significantly faster download speeds when compared to traditional ADSL connections. ADSL+2 works in

the same fashion as ADSL a special filter is installed on a subscriber's telephone line to split

existing copper telephone lines (POTS) between regular telephone (voice) and ADSL+2.

ADSL2+ service is most commonly offered in highly-populated metropolitan areas and subscribers must be in close geographical locations to the provider's central office to receive ADSL2+ service.

3. SDSL - Symmetric Digital Subscriber Line

Short for symmetric digital subscriber line, SDSL is a technology that allows more data to

be sent over existing copper telephone lines (POTS). SDSL supports data rates up to 3

Mbps. SDSL works by sending digital pulses in the high-frequency area of telephone wires

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and cannot operate simultaneously with voice connections over the same wires. SDSL requires a special SDSL modem. SDSL is called symmetric because it supports the same data rates for upstream and downstream traffic.

Connection speeds – upto 3 Mbps

4. VDSL - Very High DSL

Very High DSL (VDSL) is a DSL technology that offers fast data rates over relatively short distances – the shorter the distance, the faster the connection rate.

iii) Cable - Broadband Internet Connection

Broadband Internet connection is designed to operate over cable TV lines. Cable Internet works by using TV channel space for data transmission, with certain channels used for downstream transmission, and other channels for upstream transmission. Because the coaxial cable used by cable TV provides much greater bandwidth than telephone lines, a cable modem can be used to achieve extremely fast access.⁷ Cable providers typically implement a cap to limit capacity and accommodate more customers. Cable speeds range from 512 Kbps to 20 Mbps.

B. Mobile Internet services

i) GPRS

GPRS (General Packet Radio Service) is rapidly becoming a global standard for sending and

receiving high-speed data across the GSM network. Under GPRS one is billed by the amount

of data sent and received, in contrast to rather than conventional Circuit Switched Data services which are billed by the connection time.

The advantages of GPRS technology is that it allows users to stay connected to the Internet

under fast download speeds as no time is spent attempting to access a dial-up connection.

It allows one to stay permanently connected to the mobile Internet without running up a huge phone bill. This is because with GPRS, you are only charged for the amount of information you send and receive, not the length of time you use your phone.

This means that once one switches on their GPRS phone and connect to the mobile Internet, you can stay on all day. Similarly when one receives a call while browsing with the

GPRS service, you can simply pause your browsing session, take the call, and then carry on

reading the page you were on already.

ii) EDGE

EDGE (Enhanced Data Rates for GSM Evolution) technology represents a step between GPRS and 3G mobile systems.

EDGE allows the delivery of advanced mobile services such as the downloading of video and music clips, full multimedia messaging, high-speed colour Internet access and e-mail

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However the speed depends upon the signal quality, number of users as well as the type of

mobile phone in use. EDGE offers the following benefits:

- faster Internet access and browsing,
- faster MMS,
- faster email transfer,
- audio contents transfer,
- video contents transfer,
- games,
- as well as all other application with the same price of data transmission as with the 'classic' GPRS

iii) CDMA

Short for Code-Division Multiple Access, a digital cellular technology that uses spread spectrum techniques where a signal is spread across multiple radio frequencies resulting in

a signal with a wider bandwidth increasing its resistance to interference. Unlike competing

systems, such as GSM, that use TDMA, CDMA does not assign a specific frequency to each

user. Instead, every channel uses the full available spectrum. Individual conversations are

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encoded with a special coding scheme (where each transmitter is assigned a code) to allow

many users to share a communication medium for data and telephone communication.

CDMA consistently provides better capacity for voice and data communications than other

commercial mobile technologies, allowing more subscribers to connect at any given time,

and it is the common platform on which 3G technologies are built.

iv) HSDPA

HSDPA, short for High-Speed Downlink Packet Access, is a new protocol for mobile telephone data transmission. It is known as a 3.5G (G stands for generation) technology. Essentially, the standard provides download speeds on a mobile phone equivalent to an ADSL (Asymmetric Digital Subscriber Line) line in a home, removing any limitations placed

on the use of your phone by a slow connection. It is an evolution and improvement on W-CDMA, or Wideband Code Division Multiple Access, a 3G protocol.

HSDPA improves the data transfer rate by a factor of at least five over W-CDMA.

HSDPA can achieve theoretical data transmission speeds of 8-10 Mbps (megabits per second). Though any data can be transmitted, applications with high data demands such as

video and streaming music are the focus of HSDPA.

v) 3G

3G or Third-Generation Mobile networks are an ITU specification for the third generation of mobile communications technology. It refers to broadband digital wireless mobile telephone networks that offer increased voice capacity and provide higher data Internet Services Options rates than 2G networks. 3G can support the new range of digital

services available for mobile devices, such as internet access, multimedia applications, and

support for global roaming. 3G is capable of a bandwidth of up to 384 Kbps when a device

is stationary or moving at pedestrian speed, 128 Kbps in a car, and 2 Mbps in fixed applications. 3G will work over wireless air interfaces such as GSM, TDMA, and CDMA.

vi) 4G

4G is the fourth generation of mobile phone mobile communication technology standards. It

is a successor to 3G standard expected to address the deficiencies of 3G in terms of speed

and quality.

4G removes all barriers of speed and bandwidth and rewards every user with high stream

data.

4G puts peak speed requirements at 100 Mbps for high mobility communication (such as

from trains and cars) and 1?Gbps for low mobility communication (such as pedestrians and stationary users)

C. Wireless Internet Services

i) iBurst

iBurst is a mobile broadband wireless access system and can provide high performance, high speed, secure, mobile wireless access for business, home and office applications.

iBurst offers the following benefits:

- Up to 1 Mbps data connectivity
- Wide area wireless access means that larger cell sizes can be accommodated.
- Always on connectivity
- The service is fully compatible with existing laptops, PDAs and desktop computers - no upgrades or new devices required.
- Many characteristics of the iBurst service combine to make it more secure than existing wireless delivery modes including the directionality of the signal as well as extensive encryption of the data transferred.

ii) Wi-Fi

This is a wireless networking technology that uses radio waves to provide wireless high-speed Internet and network connections. Wi-Fi works with no physical wired connection between sender and receiver by using radio frequency (RF), a frequency within

the electromagnetic spectrum associated with radio wave propagation. When an RF

current is supplied to an antenna, an electromagnetic field is created that then is able to propagate through space. The basis of any WiFi wireless network is the access point (AP). The primary job of an access point is to broadcast a wireless signal that computers can detect and "tune" into. In order to connect to an access point and join a wireless network, computers and devices must be equipped with wireless network adapters. Usually deployed in airports, universities, bookstores, coffee shops, office campuses and private residences.

iii) WiMAX

WiMAX can be used for wireless networking in much the same way as WiFi. It is a second-generation protocol that allows for more efficient bandwidth use, interference avoidance, and is intended to allow higher data rates over longer distances.

The bandwidth and reach of WiMAX make it suitable for the following potential applications:

- Connecting Wi-Fi hotspots with each other and to other parts of the Internet.
- Providing a wireless alternative to cable and Digital Subscriber line (DSL) for last mile (last km) broadband access.
- Providing high-speed mobile data and telecommunications services.
- Providing a diverse source of Internet connectivity as part of a business continuity plan. That is, if a business has a fixed and a wireless internet connection, Internet Services Options especially from unrelated providers, they

are unlikely to be affected by the same service outage.

Shared vs dedicated bandwidth

Broadband connections can be divided into two major categories: shared and dedicated.

Shared Internet connections include DSL and Cable broadband connections. Dedicated connections are provided by leased lines such as E1, T1.

Dedicated bandwidth

Dedicated bandwidth means that the bandwidth speed that you subscribed for is committed for your exclusive use. You are guaranteed that the Bandwidth (or speed) that you signed up can always be achieved 24 hours a day regardless of total network utilization

at any given time. That means if you signed up for dedicated bandwidth of 512/512kbps,

you will always be able to receive downloads up to 512kbps if your network requires it.

Shared bandwidth

Shared bandwidth means that the bandwidth speed that you signed up for is used collectively between yourself and other users on that service plan. Your ISP will endeavour

to achieve the minimum set speed on the service level agreement but maximum speeds are

only possible when other users in your service plan are not active. This has great advantages of having the ability to have higher speeds during non peak times and when

the

network is underutilized, without having to purchase dedicated bandwidth.

Questions about the Internet

1. What is an ISP?
2. Name 4 ISPs.
3. What services do most ISPs provide?
4. What software is required to view web pages?
5. Describe 2 methods of viewing e-mail messages.
6. What is meant by the term 'dial-up' Internet access?
7. Describe 3 types of broadband access.

Practical task

1. Research the websites of 2 different Internet Service Providers (ISPs). For each one, compare the costs of
 - (a) dial-up access
 - (b) ADSL broadband access
2. Write a brief report for an older neighbour who does not have a computer, and is considering setting up Internet access. Identify the hardware and software required.
3. Recommend an ISP and outline the advantages and disadvantages of dial-up and broadband access.